

Olivia Gough

Google Scholar | github.com/omgough

PROFILE

Physicist and machine learning engineer who turns complex physical systems into predictive, decision-support tools in production code — the core of my work at Electio, where I model energy-production systems and own the optimisation recommendations delivered to customers. Condensed-matter PhD (Oxford) in the microstructure of organic semiconductors characterised by synchrotron X-ray techniques, with a record of leading large-facility beamtimes and authoring open-source research software now adopted as standard by groups at Oxford, Imperial, Sheffield, the ESRF and Lawrence Berkeley National Laboratory. The same physical-modelling, scientific-software and large-dataset expertise built in that research is what I now apply, day to day, to Electio's energy-systems products.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Apr 2026 – present

Electio — software for energy systems

Applying PhD-level physical modelling and research-software expertise to energy-production systems.

- Build physics-based models of complex energy-production systems and turn them into predictive, decision-support tools in production code — directly extending the physical-modelling and research-software work of my doctorate into industry.
- Developed a nodal-analysis and multiphase-flow model for production optimisation, and integrated its predictions as physically-grounded features into a machine-learning model — the same modelling-plus-data-pipeline discipline I developed on large *in situ* synchrotron datasets.
- Own the optimisation recommendations delivered to customers (e.g. gas-injection strategy), derived from historical operating data and model output — accountable for the technical advice given.
- Bring specialist scientific-software practice (tested, documented, maintainable packages) into the production codebase, having previously authored facility-grade tools used across international research groups (see Software).

SOFTWARE

giwaxs-analysis

Open-source Python package — github.com/omgough/giwaxs-analysis

- Python tools for GIWAXS data processing built on pyFAI and fabio: calibration, 1D/sector/cake integration, *in situ* heatmaps, peak fitting and Scherrer crystallite-size analysis.
- Properly engineered and maintained — packaged (`pyproject.toml`, conda environment), tested, documented and MIT-licensed, with a `CITATION.cff` for citable reuse.
- Adopted as standard by groups working on *in situ* GIWAXS: Riede group (Oxford), Heutz group (Imperial), University of Sheffield, ESRF, and the Su Materials Lab (Lawrence Berkeley National Laboratory).

giwaxs_toolbox

Contributor / co-author — official Diamond Light Source toolbox for beamline I07 github.com/DiamondLightSource/giwaxs_toolbox

- Contributor to the Diamond Light Source GIWAXS toolbox: Python tools for visualising and analysing GIWAXS data collected on beamline I07, designed to operate on the output of the `fast_rsm` mapping package.
- Interactive widgets for exploring *in situ* GIWAXS datasets and extracting 1D profiles for further analysis, including peak fitting.
- Maintained, facility-grade package (Apache-2.0): src-layout with `pyproject.toml`, unit tests, GitHub Actions CI, Sphinx documentation with walkthroughs and training videos, and containerised (Docker) deployment on the Diamond filesystem for beamline users.

RESEARCH EXPERIENCE

PhD, Condensed Matter Physics

Oct 2021 – Dec 2025

University of Oxford — supervisor Prof. Moritz Riede

In situ microstructural characterisation of organic solar cells.

- Investigated the relationship between microstructure, optoelectronic properties and device performance in organic photovoltaics.
- Investigative lead on 8 successful beamtime proposals (100% acceptance) across multiple international facilities, with full responsibility for experiment planning, execution and analysis.
- Expert in synchrotron / large-facility characterisation (GIWAXS, NEXAFS, RSoXS) and in thin-film processing by vacuum thermal evaporation.

- Built a Python data-analysis pipeline for large *in situ* GIWAXS datasets (pandas, pyFAI, pygix, fabio, matplotlib), now used as standard by other beamtime users.
- Proficient with in-house techniques (AFM, UV-Vis spectroscopy) and large-dataset handling over Globus and FileZilla.

MRes, Plastic Electronics

Oct 2020 – Sep 2021

Imperial College London — supervisor Prof. Jenny Nelson

Distinction

Polymer battery electrodes and side-chain engineering.

- Characterised a set of conjugated polymers with varying polar side chains for charge-storage devices using cyclic voltammetry, spectroelectrochemistry and galvanostatic charge/discharge.
- Related battery-system performance to molecular design; thin-film solution processing by blade coating and electrochemical half-cells.

BSc Research Project

2019 – 2020

Queen Mary University of London — supervisor Dr. Theo Kreouzis

- Fabricated Schottky diodes by solution processing and measured hole mobility in polythiophenes by the time-of-flight technique.

LARGE-FACILITY & SYNCHROTRON EXPERIENCE

Advanced Light Source, LBNL, USA

Doctoral Fellowship (two consecutive years)

- Won a full-time ALS Doctoral Fellowship in Residence two years running; successful Molecular Foundry proposal for sample fabrication.
- NEXAFS and RSoXS at beamline 11.0.1.2; GIWAXS at beamline 7.3.3.

Diamond Light Source, UK

Lead investigator / superuser — *in situ* GIWAXS, MINERVA chamber at I07

- Operated the MINERVA vacuum chamber and the scattering experiment simultaneously — a demanding combined technique.
- Superuser of the Block Allocation Group: responsible for training new users and coordinating two weeks of beamtime per cycle.
- Authored usage documentation and the Python analysis scripts now adopted by the group (see Research Software).

Other facilities

- European Synchrotron Radiation Facility, France — *ex situ* GIWAXS at XMaS.
- ISIS Neutron and Muon Source, UK — small-angle neutron scattering at Larmor.
- Diamond Light Source, UK — X-ray reflectivity at I07.

TECHNICAL SKILLS

Programming & modelling: Python (pandas, NumPy, matplotlib, pyFAI, pygix, fabio), data-analysis pipelines, physical modelling, machine learning, nodal / multiphase-flow analysis.

Experimental techniques: GIWAXS, NEXAFS, RSoXS, GISAXS, small-angle neutron scattering, X-ray reflectivity, AFM, UV-Vis spectroscopy, cyclic voltammetry; thin-film growth by vacuum thermal evaporation and solution processing.

Tools & infrastructure: git/GitHub, Jupyter, conda, large-dataset transfer (Globus, FileZilla), $\LaTeX$.

SELECTED PUBLICATIONS & PRESENTATIONS

Peer-reviewed journal articles

1. Zhu, M., Tsokkou, D., **Gough, O.**, Trinkaus, K., Evans, N., Kaienburg, P., Meerholz, K., Olthof, S., Banerji, N. & Riede, M., “Low-voltage-loss vacuum thermally evaporated BHJ organic solar cells with DCV3T as non-fullerene acceptor”, *Journal of Physics D: Applied Physics* (accepted, in press, 2026).
2. **Gough, O.**, Trinkaus, K., Kaienburg, P., Li, Z., Lauritzen, A.E., Rawle, J., Norris, H., Hilfiker, J., Smith, J., Veneri, A., Su, G.M. & Riede, M., “*In situ* growth studies of vacuum-thermally evaporated films using ellagic acid templating for organic photovoltaics”, *Organic Electronics* **156**, 107445 (2026). [\[link\]](#)
3. Yu, H., Marks, A., Tuladhar, S.M., Siemons, N., Anderson, I., Bidinger, S., Keene, S.T., Quill, T.J., Wu, R., **Gough, O.**, *et al.*, “The influence of alkyl spacers and molecular weight on the charge transport and storage properties of oxybithiophene-based conjugated polymers”, *Angewandte Chemie International Edition* **64**, e202417897 (2025). [\[link\]](#)
4. Ramadan, A.J., Jeong, W.H., Oliver, R.D.J., Jiang, J., Dasgupta, A., Yuan, Z., Smith, J., Lee, J.E., Motti, S.G., **Gough, O.**, *et al.*, “The role of the organic cation in developing efficient green perovskite LEDs based on quasi-2D perovskite heterostructures”, *Advanced Functional Materials* (2024).

5. Habib, I., Kaienburg, P., Xia, D., **Gough, O.**, *et al.*, “Vacuum-deposited organic solar cells with BTIC-H as an A–D–A non-fullerene acceptor”, *APL Materials* (2023).

Under review / in preparation

1. **Gough, O.**, Lauritzen, A.E., Kaienburg, P., Dalglish, R. & Riede, M., “Small-angle neutron scattering to evaluate the impact of annealing on small-molecule bulk heterojunctions for organic photovoltaics”, manuscript in preparation (2025).

Conference presentations

1. **Gough, O.**, Trinkaus, K., Kaienburg, P., Su, G. & Riede, M., “*In situ* growth observations of vacuum thermally evaporated films with ellagic acid as a templating layer for organic photovoltaics”, MRS Spring (2025).
2. **Gough, O.**, Trinkaus, K., Kaienburg, P., Su, G. & Riede, M., “Tuning the microstructure of organic photovoltaics using rylene-dye-derivative templating layers with X-ray characterisation techniques”, MRS Spring (2025).
3. **Gough, O.**, Lauritzen, A.E., Li, Z., Huss-Hansen, M. & Riede, M., “*In situ* microstructural observations of vacuum-thermally-evaporated organic thin-film growth with ellagic acid as a templating layer”, SNI22, Berlin (2022).

TEACHING, SUPERVISION & OUTREACH

- Lab demonstrator for undergraduate Thermal Physics and Condensed Matter Physics modules; taught a dye-sensitised solar cell (DSSC) workshop for MSc Energy Systems students using Caltech material.
- Supervised an MSc project using GIWAXS to study the effect of templating layers on microstructure.
- Designed and ran a day-long DSSC workshop for 14–16-year-old girls at St Paul’s Way Trust School, London (2023), encouraging further study of physics.
- Workshop leader, Marie Curious outreach event, University of Oxford (2023, ages 11–14); group leader, Marie Curious at Kings High School, Warwick (2022).

AWARDS & FELLOWSHIPS

- Advanced Light Source Doctoral Fellowship in Residence, LBNL — awarded two consecutive years.
- Oxford–Wolfson–Marriott–Condensed Matter Physics Graduate Scholarship.
- UKRI doctoral studentship.

EDUCATION

PhD, Condensed Matter Physics

University of Oxford

Oct 2021 – Dec 2025

MRes, Plastic Electronics

Imperial College London

Oct 2020 – Sep 2021

Distinction

BSc, Physics

Queen Mary University of London

Sep 2017 – Jun 2020

First Class Honours

LANGUAGES

Native English; limited working proficiency in Italian.